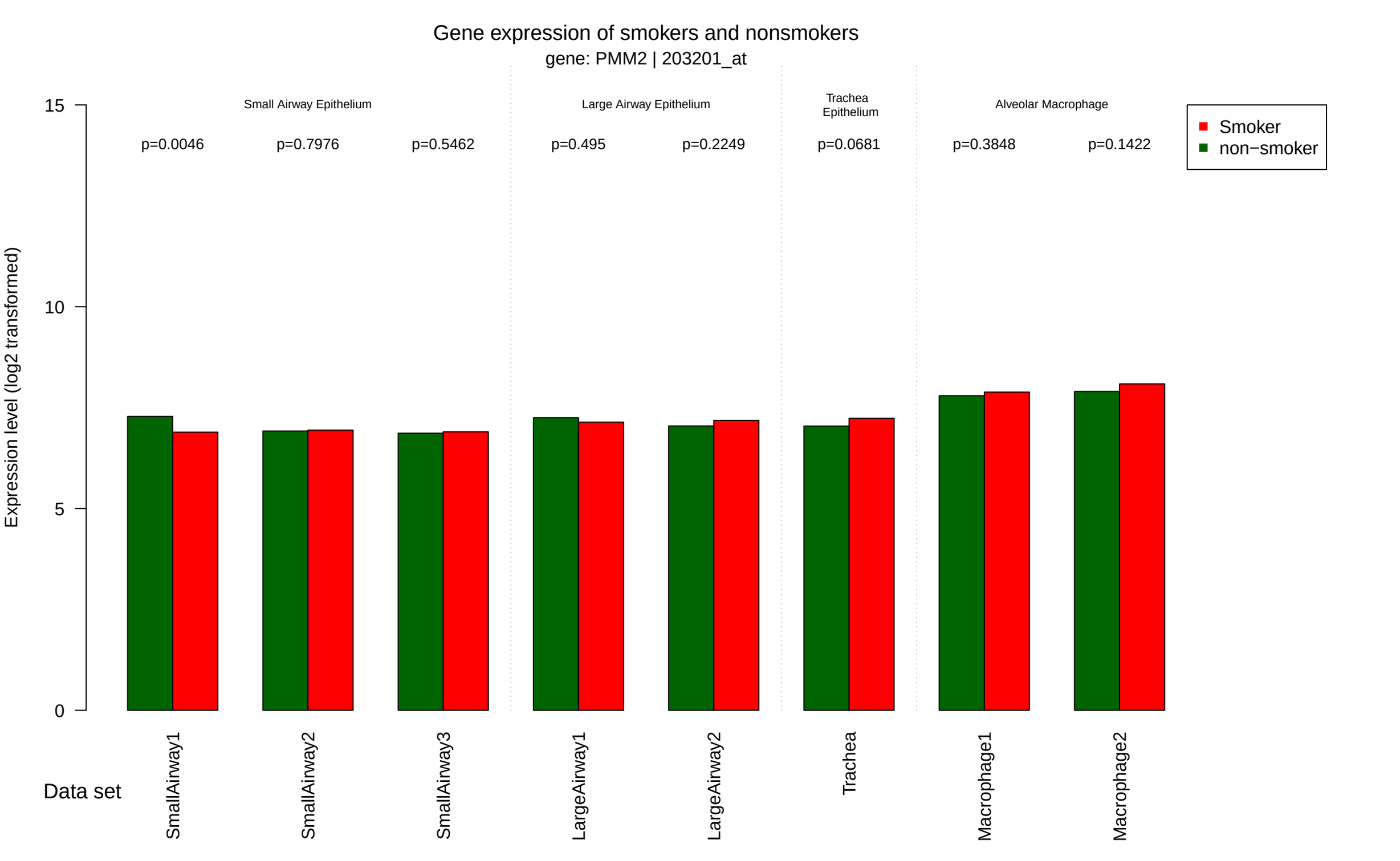
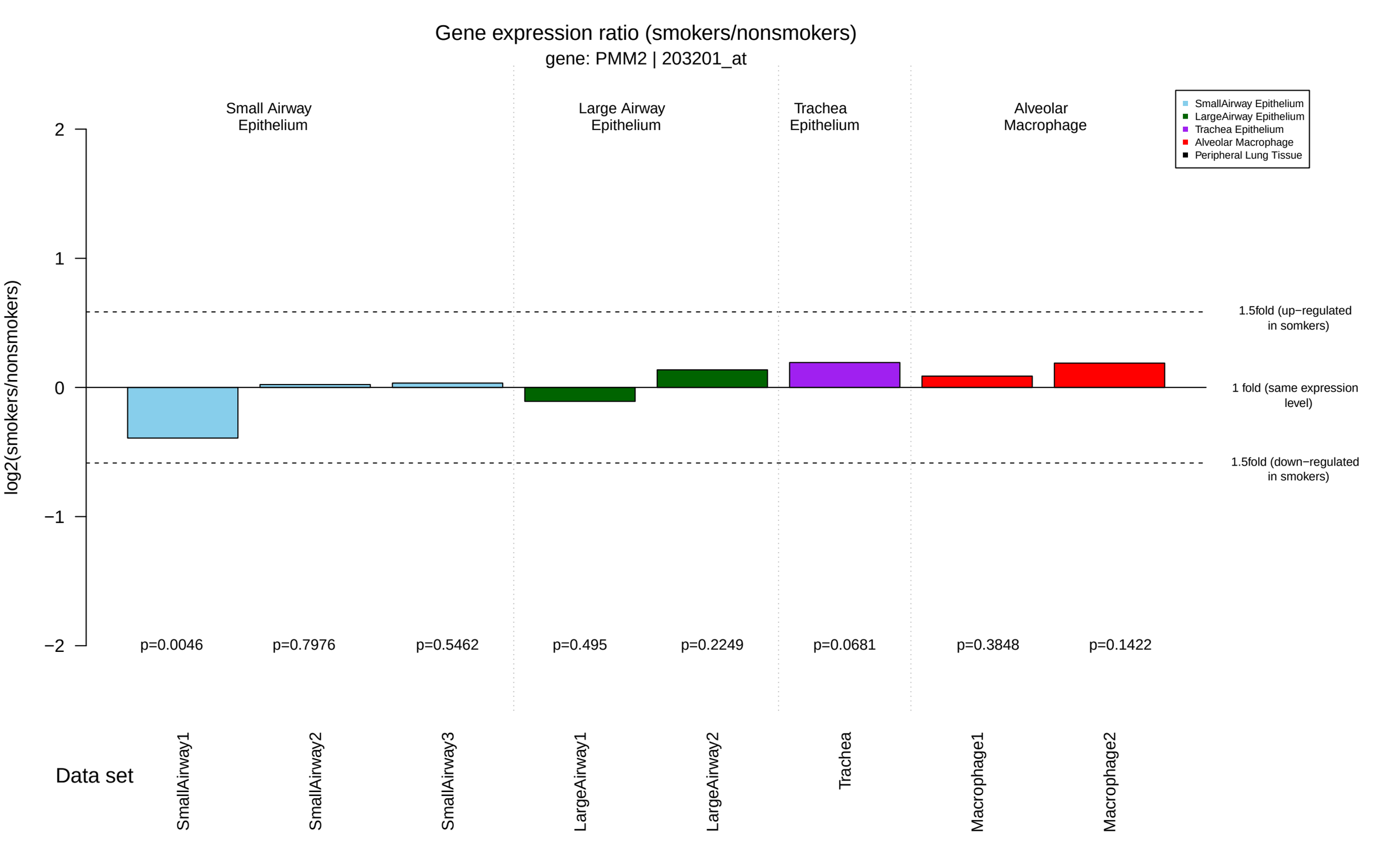
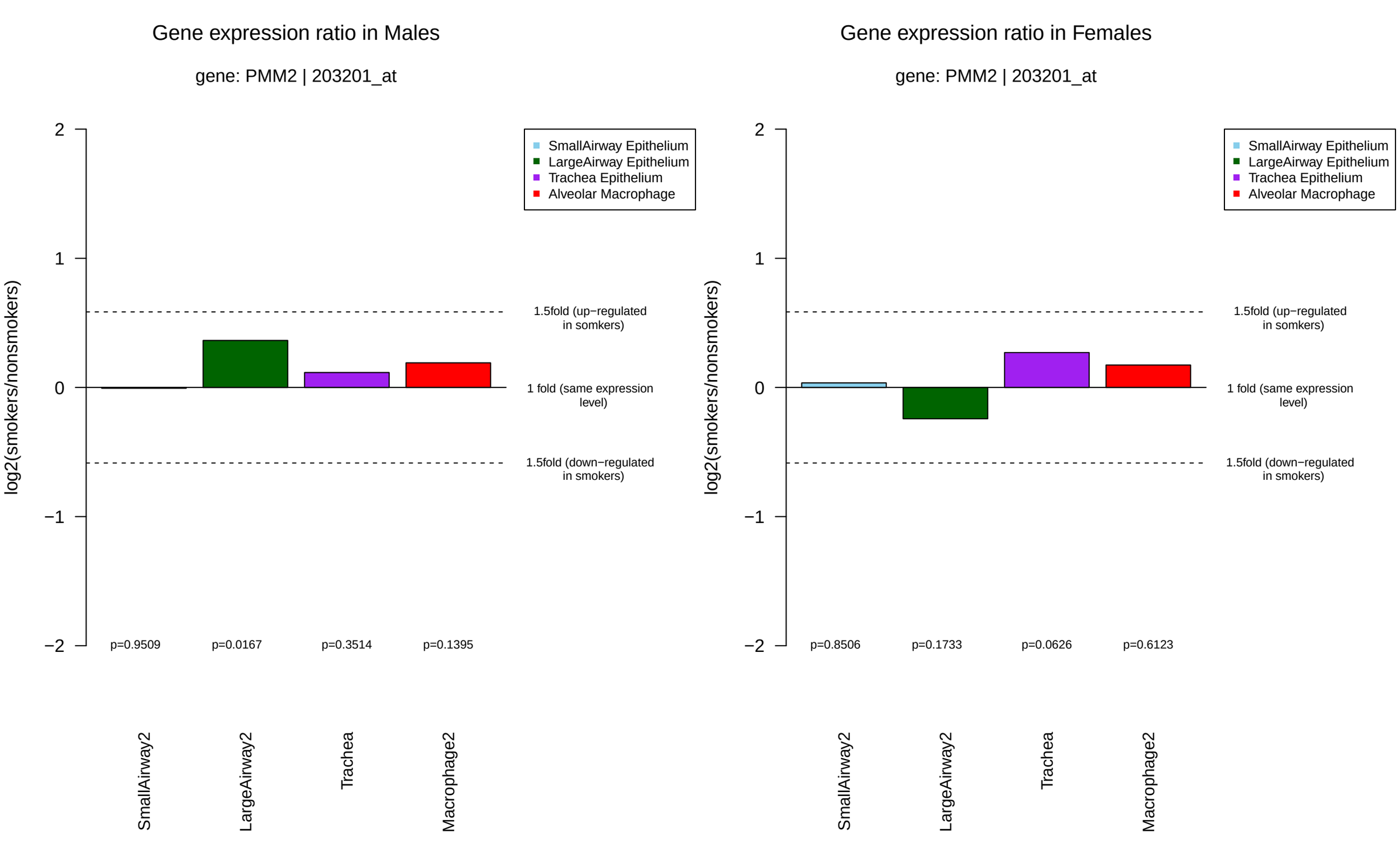
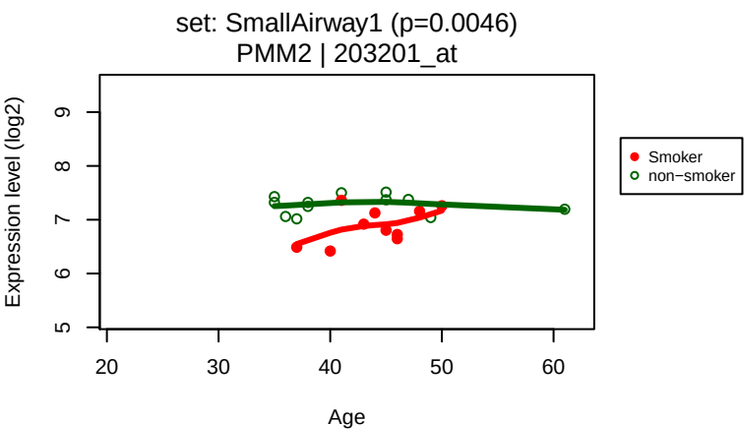
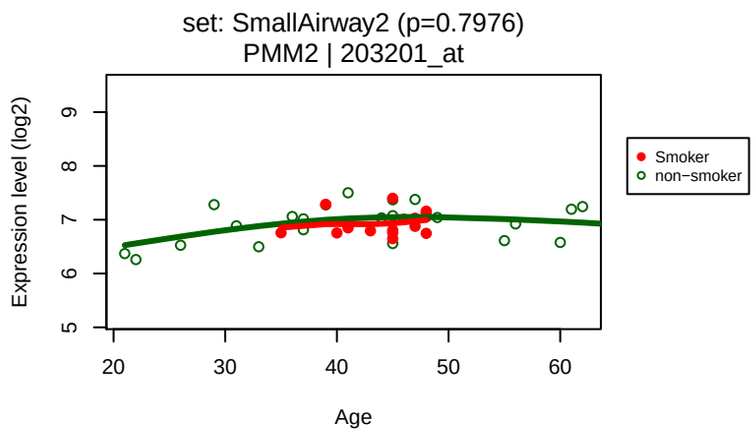
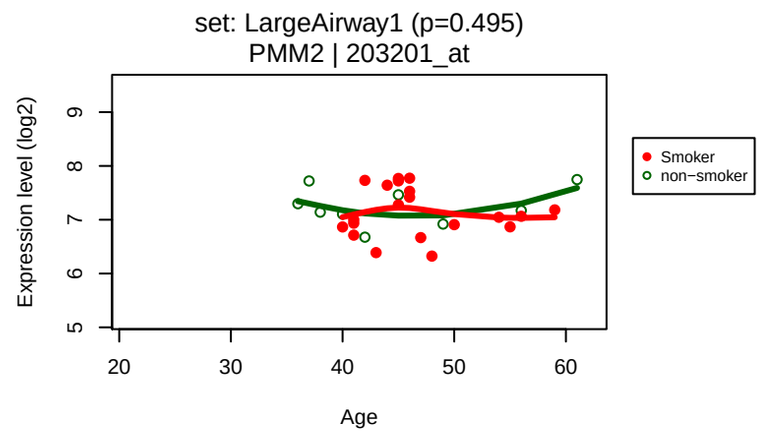

Gene expression of smokers and nonsmokers
gene: PMM2 | 203201_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0046
p=0.7976
p=0.5462
p=0.495
p=0.2249
p=0.0681
p=0.3848
p=0.1422
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

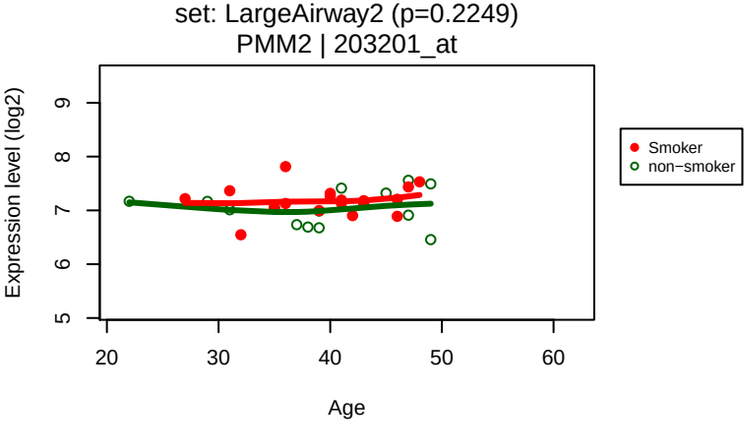

Gene expression ratio (smokers/nonsmokers)
gene: PMM2 | 203201_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0046
p=0.7976
p=0.5462
p=0.495
p=0.2249
p=0.0681
p=0.3848
p=0.1422
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

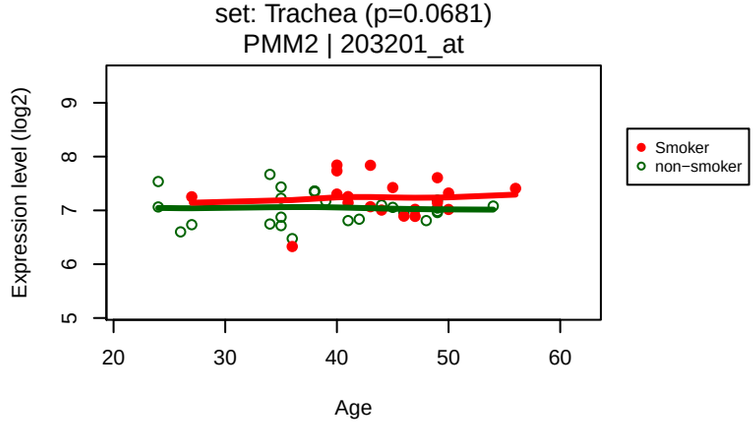

Gene expression ratio in Males
gene: PMM2 | 203201_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9509
p=0.0167
p=0.3514
p=0.1395
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PMM2 | 203201_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8506
p=0.1733
p=0.0626
p=0.6123
SmallAirway2
LargeAirway2
Trachea
Macrophage2

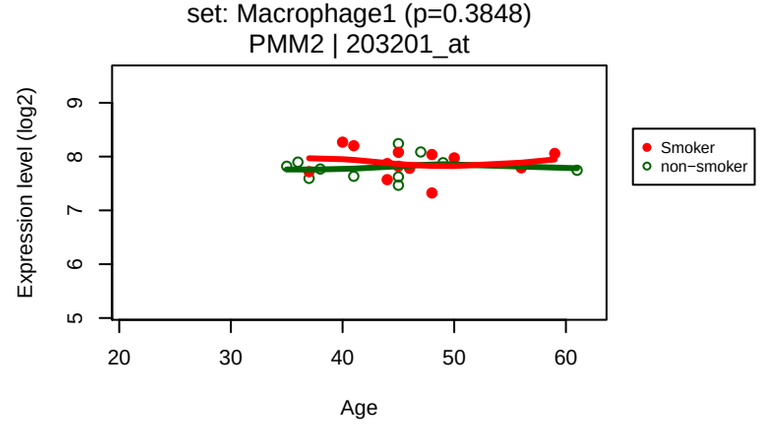

set: SmallAirway1 (p=0.0046)
PMM2 | 203201_at
Expression level (log2)
Age
Smoker
non−smoker

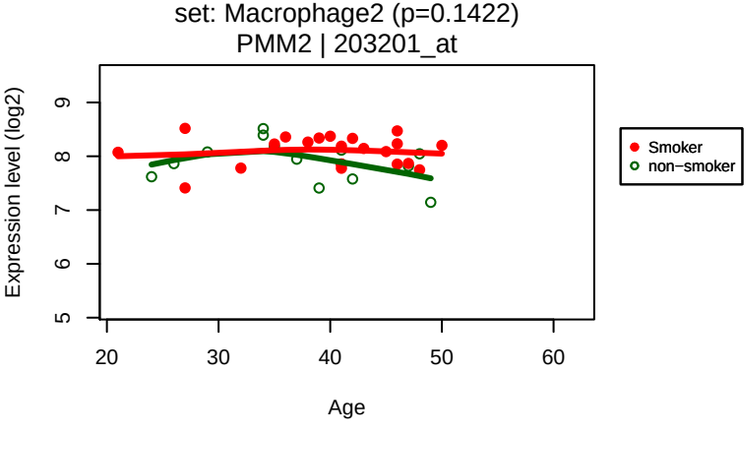

set: SmallAirway2 (p=0.7976)
PMM2 | 203201_at
Expression level (log2)
Age
Smoker
non−smoker

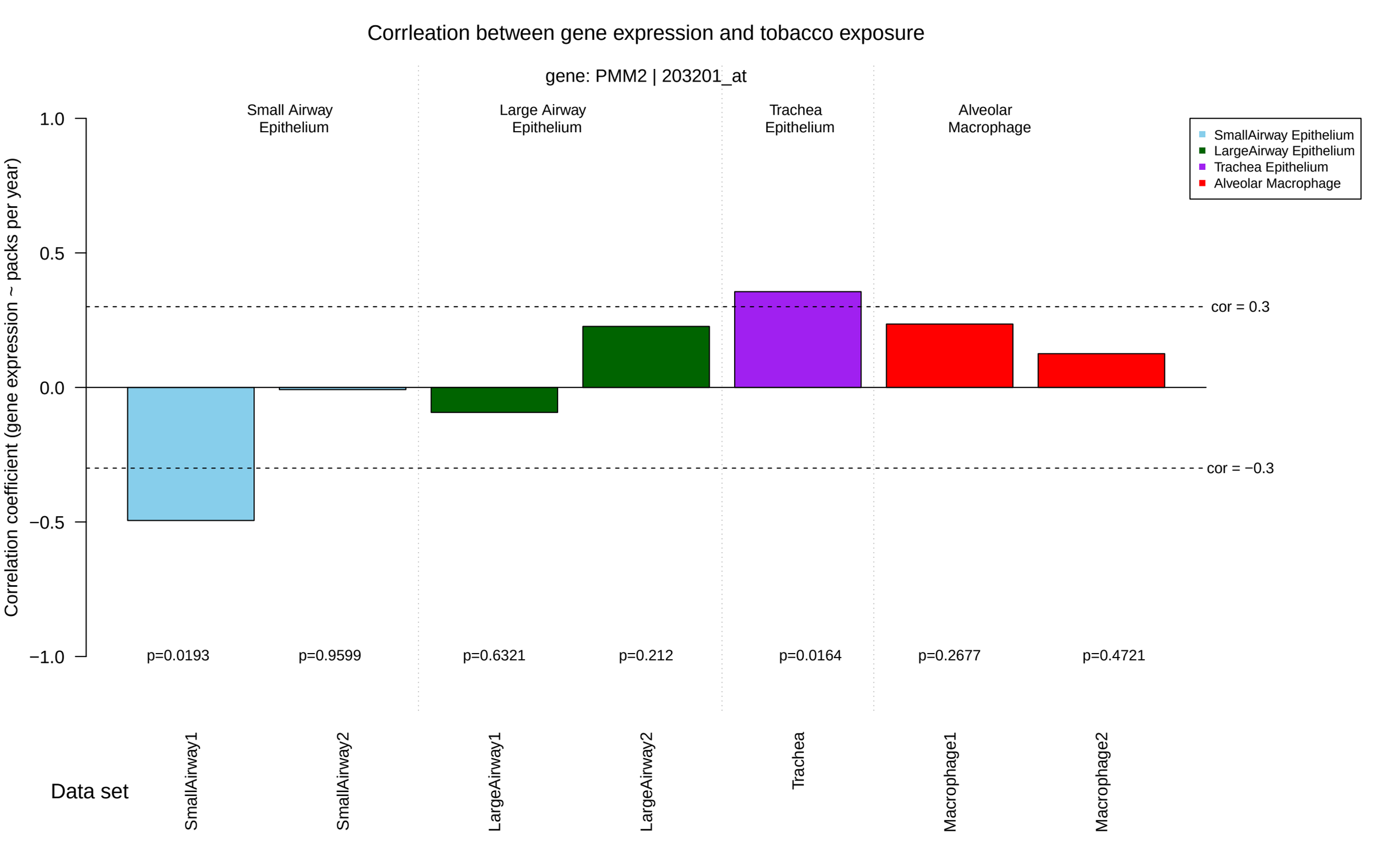

set: LargeAirway1 (p=0.495)
PMM2 | 203201_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2249)
PMM2 | 203201_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0681)
PMM2 | 203201_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3848)
PMM2 | 203201_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1422)
PMM2 | 203201_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PMM2 | 203201_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0193
p=0.9599
p=0.6321
p=0.212
p=0.0164
p=0.2677
p=0.4721
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2